

List the enzymes used in genetic engineering (genetic modification) **and** outline their roles in natural processes.

(b) Genetically modified organisms (GMOs) include crop plants. Genetically modified (GM) crop plants have been grown in North America in increasing quantities since 1996. They are now grown in many areas of the world and are eaten by millions of people and farm animals.

Table 4.1

continent	area of land used to grow GM crop plants / million hectares	edible GM crop plants grown
Africa	3.5	maize, soybean
Asia	19.1	maize, papaya, tomato
Australasia	0.6	canola
Europe	0.2	maize
North America	81.1	canola, maize, papaya, soybean, squash, sugar beet
South America	70.9	canola, maize, soybean

- (i) Suggest how North America and South America benefit from growing large areas of GM crop plants rather than non-GM crop plants.

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- (ii) Only a small area of land is used to grow GM crop plants in Europe. This is mainly because most of Europe uses a 'precautionary principle'. The precautionary principle prevents GM crop plants being grown if there is a possible risk of harm to human health or to the environment, even if there is no proof of harm.

Many European citizens have concerns about the safety of eating GM food, but Europe imports large quantities of GM maize and GM soybean to feed farm animals.

Use this information and Table 4.1 to deduce two arguments **in favour of** growing more GM crop plants in Europe.

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- (iii) Explain why the data in Table 4.1 are not enough to calculate the extent to which different continents have replaced traditional crop plants with GM versions.

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